

PMP Formulas Cheat Sheet

Earned Value Management (EVM)

Core Values

Formula	Calculation	Meaning
PV (Planned Value)	Budgeted cost of work scheduled	What you <i>planned</i> to spend
EV (Earned Value)	Budgeted cost of work performed	Value of work <i>actually done</i>
AC (Actual Cost)	Actual cost of work performed	What you <i>actually spent</i>
BAC (Budget at Completion)	Total project budget	Original total budget

Variance Analysis

Formula	Calculation	Interpretation
CV (Cost Variance)	$EV - AC$	Negative = over budget
SV (Schedule Variance)	$EV - PV$	Negative = behind schedule

Performance Indices

Formula	Calculation	Interpretation
CPI (Cost Performance Index)	$EV \div AC$	<1 = over budget
SPI (Schedule Performance Index)	$EV \div PV$	<1 = behind schedule

Memory trick: Variance = EV minus something, Index = EV divided by something

Forecasting

Formula	Calculation	When to Use
EAC (Estimate at Completion)	$BAC \div CPI$	Typical (current CPI continues)

Formula	Calculation	When to Use
EAC	$AC + (BAC - EV)$	Atypical (original estimates valid)
EAC	$AC + \text{Bottom-up ETC}$	Re-estimate remaining work
EAC	$AC + [(BAC - EV) \div (CPI \times SPI)]$	Both cost & schedule matter
ETC (Estimate to Complete)	$EAC - AC$	How much more to spend
VAC (Variance at Completion)	$BAC - EAC$	Expected variance at end
TCPI (To-Complete Performance Index)	$(BAC - EV) \div (BAC - AC)$	Required efficiency to meet BAC
TCPI	$(BAC - EV) \div (EAC - AC)$	Required efficiency to meet EAC

PERT (Three-Point Estimating)

Formula	Calculation
Expected Duration (Triangular)	$(O + M + P) \div 3$
Expected Duration (Beta/PERT)	$(O + 4M + P) \div 6$
Standard Deviation (σ)	$(P - O) \div 6$
Variance	$\sigma^2 = [(P - O) \div 6]^2$

O = Optimistic, M = Most Likely, P = Pessimistic

Confidence Levels

- $\pm 1\sigma = 68.26\%$
- $\pm 2\sigma = 95.46\%$
- $\pm 3\sigma = 99.73\%$
- $\pm 6\sigma = 99.99\%$

Communication Channels

Formula	Calculation
Channels	$n(n - 1) \div 2$

n = number of stakeholders

Quick reference: 5 people = 10 channels, 10 people = 45 channels

Network Diagram / Critical Path

Term	Calculation
Early Start (ES)	Max(EF of predecessors), or 0 if first
Early Finish (EF)	ES + Duration
Late Finish (LF)	Min(LS of successors), or EF if last
Late Start (LS)	LF – Duration
Total Float	LS – ES <i>or</i> LF – EF
Free Float	ES(successor) – EF(current)

Critical Path: Longest path through the network; Total Float = 0

Financial Formulas

Formula	Calculation	Use
Present Value (PV)	$FV \div (1 + r)^n$	Value today of future money
Future Value (FV)	$PV \times (1 + r)^n$	Value of money in future
Net Present Value (NPV)	$\Sigma(\text{Cash flows} \div (1+r)^n) - \text{Initial}$	Higher = better project

Formula	Calculation	Use
ROI	$(\text{Gain} - \text{Cost}) \div \text{Cost} \times 100$	Percentage return
Benefit-Cost Ratio (BCR)	$\text{Benefits} \div \text{Costs}$	>1 = worthwhile
Payback Period	$\text{Initial Investment} \div \text{Periodic Cash Flow}$	Time to recover costs

$r = \text{interest rate}, n = \text{number of periods}$

Procurement

Formula	Calculation
Point of Total Assumption (PTA)	$[(\text{Ceiling} - \text{Target Price}) \div \text{Buyer Share \%}] + \text{Target Cost}$

Used in Fixed Price Incentive Fee (FPIF) contracts

Depreciation

Method	Calculation
Straight Line	$(\text{Asset Cost} - \text{Salvage Value}) \div \text{Useful Life}$
Double Declining Balance	$(2 \div \text{Useful Life}) \times \text{Book Value}$

Quick Reference Card

Question	Formula
Are we over/under budget?	$\text{CV} = \text{EV} - \text{AC}$
Are we ahead/behind schedule?	$\text{SV} = \text{EV} - \text{PV}$
How efficiently are we using money?	$\text{CPI} = \text{EV} \div \text{AC}$

Question	Formula
How efficiently are we using time?	$SPI = EV \div PV$
What will the project cost?	$EAC = BAC \div CPI$
How much more will we spend?	$ETC = EAC - AC$
How many communication paths?	$n(n-1) \div 2$
Expected duration (PERT)?	$(O + 4M + P) \div 6$

